CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

51. Identify the inorganic sulphides that are yellow in colour:

(A) $(NH_4)_2S$ (B) PbS

(C) CuS (D) As_2S_3

(E) As_2S_5

Choose the correct answer from the options given below:

(1) (A), (D) and (E) only

(2) (A) and (B) only

(3) (A) and (C) only

(4) (D) and (E) only

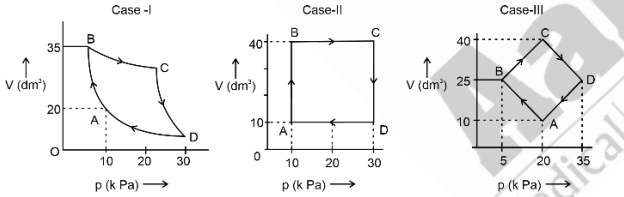
Answer (1)

Sol. $(NH_4)_2S$: colourless to yellow

PbS and CuS : Black

As_2S_3 and As_2S_5 : Yellow

52.



An ideal gas undergoes a cyclic transformation starting from the point A and coming back to the same point by tracing the path A→B→C→D→A as shown in the three cases above.

Choose the correct option regarding ΔU:

(1) ΔU(Case-I) = ΔU(Case-II) = ΔU(Case-III)

(2) ΔU(Case-III) > ΔU(Case-II) > ΔU(Case-I)

(3) ΔU(Case-I) > ΔU(Case-III) > ΔU(Case-II)

(4) ΔU(Case-I) > ΔU(Case-II) > ΔU(Case-III)

Answer (1)

Sol. As all the three case are cyclic process. Change in state functions will be zero.

Hence, $\Delta U = 0$ for all.

53. Concentrated nitric acid is labelled as 75% by mass. The volume in mL of the solution which contains 30 g of nitric acid is _____.

Given: Density of nitric acid solution is 1.25 g/mL.

(1) 55 (2) 32

(3) 45 (4) 40

Answer (2)

Sol. 75% by mass means :

$\because$ 75 g HNO_3 in 100 g solution.

$\therefore$ 30 g HNO_3 in $\frac{100}{75}\times 30$ g solution.

$$m_{solution} = \frac{100\times 30}{75}\ g$$

$$\because\ V_{sol} = \frac{m_{sol}}{d_{sol}} = \frac{100\times 30}{75\times 1.25} = 32\ mL$$

54. Identify correct conversion during acidic hydrolysis from the following:

(A) starch gives galactose.

(B) cane sugar gives equal amount of glucose and fructose.

(C) milk sugar gives glucose and galactose.

(D) amylopectin gives glucose and fructose.

(E) amylose gives only glucose.

Choose the **correct** answer from the option given below

(1) (B), (C) and (D) only (2) (A), (B) and (C) only

(3) (C), (D) and (E) only (4) (B), (C) and (E) only

Answer (4)

Sol. Starch gives glucose on hydrolysis amylose and amylopectin are components of starch.

Hence, (B), (C) and (E) are correct.



55. Identify product [A], [B] and [C] in the following reaction sequence.

$$CH_3-C\equiv CH \xrightarrow[H_2]{Pd/C} [A] \xrightarrow[(ii)\ Zn,\ H_2O]{(i)\ O_3} [B]+[C]$$

(1) [A] : $CH_3CH_2CH_3$, [B] : CH_3CHO, [C] : HCHO

(2) [A] : $CH_2 = CH_2$, [B] : $H_3C-\overset{\overset{O}{\|}}{C}-CH_3$, [C] : HCHO

(3) [A] : $CH_3 - CH = CH_2$, [B] : CH_3CHO, [C] : HCHO

(4) [A] : $CH_3 - CH = CH_2$, [B] : CH_3CHO, [C] : CH_3CH_2OH

Answer (3)

Sol. $CH_3-C\equiv CH \xrightarrow[H_2]{Pd/C} CH_3-CH=CH_2$

$$\xrightarrow[(ii)\ Zn,\ H_2O]{(i)\ O_3} CH_3-CHO+HCHO$$

56. Given below are two statements:

Statement (I): According to the Law of Octaves, the elements were arranged in the increasing order of their atomic number.

Statement (II): Meyer observed a periodically repeated pattern upon plotting physical properties of certain elements against their respective atomic numbers.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are true

(2) Statement I is false but Statement II is true

(3) Statement I is true but Statement II is false

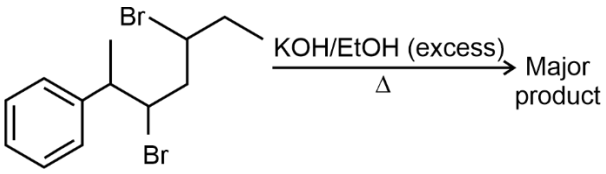
(4) Both Statement I and Statement II are false

Answer (4)

Sol. In Law of Octaves, atomic mass is used for arrangement.

Mayer observed the pattern against atomic mass.

57. The major product of the following reaction is :



(1) 2-Phenylhepta-2,4-diene

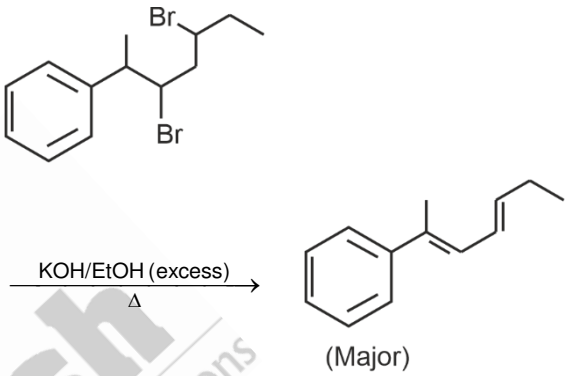
(2) 6-Phenylhepta-3,5-diene

(3) 6-Phenylhepta-2,4-diene

(4) 2-Phenylhepta-2,5-diene

Answer (1)

Sol.



58. Which of the following is/are not correct with respect to energy of atomic orbitals of hydrogen atom?

(A) 1s < 2p < 3d < 4s

(B) 1s < 2s = 2p < 3s = 3p

(C) 1s < 2s < 2p < 3s < 3p

(D) 1s < 2s < 4s < 3d

Choose the **correct** answer from the options given below:

(1) (B) and (D) only (2) (A) and (C) only

(3) (C) and (D) only (4) (A) and (B) only

Answer (3)

Sol. For H-atom:

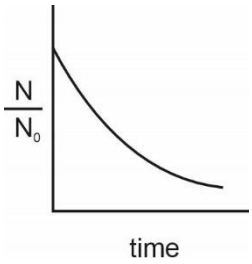
Energy of $1s < 2s = 2p < 3s = 3p = 3d < 4s = 4p = 4d = 4f$.

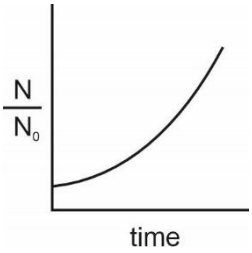
Hence (C) and (D) are incorrect.

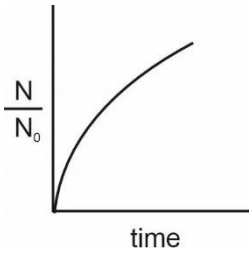


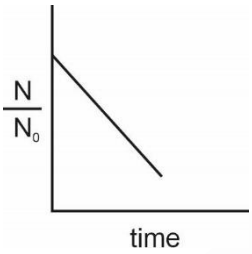
59. For bacterial growth in a cell culture, growth law is very similar to the law of radioactive decay. Which of the following graphs is most suitable to represent bacterial colony growth?

Where N – Number of Bacteria at any time, N_0 – Initial number of Bacteria.

(1) 

(2) 

(3) 

(4) 

Answer (2)

Sol. For Bacterial growth following 1st order:

$\frac{N}{N_0}$ will increase with time exponentially.

60. Arrange the following in increasing order of solubility product: $Ca(OH)_2$, AgBr, PbS, HgS

(1) HgS < PbS < AgBr < $Ca(OH)_2$

(2) PbS < HgS < $Ca(OH)_2$ < AgBr

(3) $Ca(OH)_2$ < AgBr < HgS < PbS

(4) HgS < AgBr < PbS < $Ca(OH)_2$

Answer (1)

Sol. Solubility of HgS < PbS < AgBr < $Ca(OH)_2$.

61. Match **List – I** with **List – II.**

	List – I (Complex)		**List – I (Hybridisation of central metal ion)**
(A)	$[CoF_6]^{3-}$	(I)	d^2sp^3
(B)	$[NiCl_4]^{2-}$	(II)	sp^3
(C)	$[Co(NH_3)_6]^{3+}$	(III)	sp^3d^2
(D)	$[Ni(CN)_4]^{2-}$	(IV)	dsp^2

Choose the correct answer from the options given below:

(1) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

(2) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

(3) (A)-(I), (B)-(IV), (C)-(III), (D)-(II)

(4) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

Answer (2)

Sol. $[CoF_6]^{3-}$: Co^{3+} with WFL – sp^3d^2

$[NiCl_4]^{2-}$: Ni^{2+} with WFL – sp^3

$[Co(NH_3)_6]^{3+}$: Co^{3+} with SFL – d^2sp^3

$[Ni(CN)_4]^{2-}$: Ni^{2+} with SFL – dsp^2

62. Assume a living cell with 0.9%(w/w) of glucose solution (aqueous). This cell is immersed in another solution having equal mole fraction of glucose and water.

(Consider the data upto first decimal place only)

The cell will :

(1) Shrink since solution is 0.45%(w/w) as a result of association of glucose molecules (due to hydrogen bonding)

(2) Show no change in volume since solution is 0.9% w/w

(3) Shrink since solution is 0.5% (w/w)

(4) Swell up since solution in 1%(w/w)

Answer (No option is correct)

Sol. Living cell has 0.9 g glucose in 100 g solution.

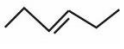
Which is dipped in a solution with $X_{H_2O} = \frac{1}{2}$.

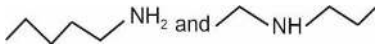
Weight of solution = $\frac{1}{2} \times 180 + \frac{1}{2} \times 18 = 99$ g.

So, the solution has 90 g glucose in 99 g solution or % w/w = 90.9%



63. Given below are two statements:

Statement (I) :  and  are isomeric compounds.

Statement (II) :  are functional group isomers.

In the light of the above statements, choose the **correct** answer from the options given below:

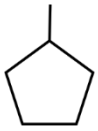
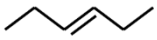
(1) **Statement I** is false but **Statement II** is true

(2) **Statement I** is true but **Statement II** is false

(3) Both **Statement I** and **Statement II** are false

(4) Both **Statement I** and **Statement II** are true

Answer (4)

Sol.  and  are ring chain isomers.

Primary and secondary amines are functional group isomers.

64. Consider an elementary reaction

$A(g) + B(g) \rightarrow C(g) + D(g)$

If the volume of reaction mixture is suddenly reduced to $\frac{1}{3}$ of its initial volume, the reaction rate will become 'x' times of the original reaction rate. The value of x is :

(1) $\frac{1}{9}$ (2) 9

(3) 3 (4) $\frac{1}{3}$

Answer (2)

Sol. Since, the reaction is elementary

$Rate = k[A]^1[B]^1$

When V is reduced to $\frac{1}{3}V$, concentration will be tripled.

Hence, $rate = 9 \times (rate)_{initial}$

$x = 9$

65. Identify correct statements:

(A) Primary amines do not give diazonium salts when treated with $NaNO_2$ in acidic condition.

(B) Aliphatic and aromatic primary amines on heating with $CHCl_3$ and ethanolic KOH form carbylamines.

(C) Secondary and tertiary amines also give carbylamine test.

(D) Benzenesulfonyl chloride is known as Hinsberg's reagent.

(E) Tertiary amines reacts with benzenesulfonyl chloride very easily.

Choose the **correct** answer from the options given below:

(1) (D) and (E) only (2) (A) and (B) only

(3) (B) and (D) only (4) (B) and (C) only

Answer (3)

Sol. - Primary aromatic amines gives diazonium salt when treated with $NaNO_2$ in acidic medium.

- Only primary amines give carbylamine test.
- Tertiary amines do not react with benzenesulfonyl chloride.

66. The purification method based on the following physical transformation is:

$$\underset{(X)}{\text{Solid}} \xrightarrow{\text{Heat}} \underset{(X)}{\text{Vapour}} \xrightarrow{\text{Cool}} \underset{(X)}{\text{Solid}}$$

(1) Crystallization (2) Sublimation

(3) Extraction (4) Distillation

Answer (2)

Sol. Phase transfer from solid to vapour directly is known as sublimation.

67. The amphoteric oxide among V_2O_3, V_2O_4 and V_2O_5, upon reaction with alkali leads to formation of an oxide anion. The oxidation state of V in the oxide anion is :

(1) +7 (2)+3

(3) +4 (4) +5

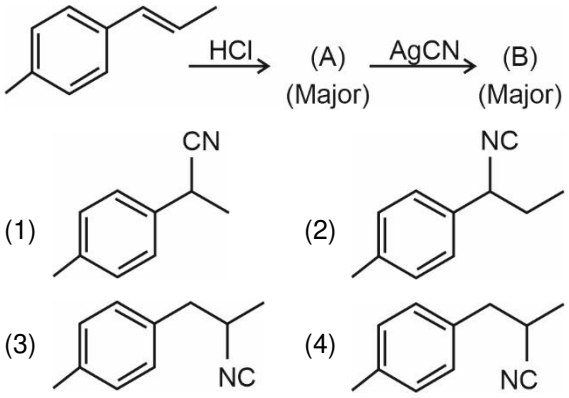
Answer (4)

Sol. V_2O_5 is amphoteric oxide.

V_2O_5 gives VO_4^{3-} on reaction with alkali oxidation state of V in VO_4^{3-} = +5

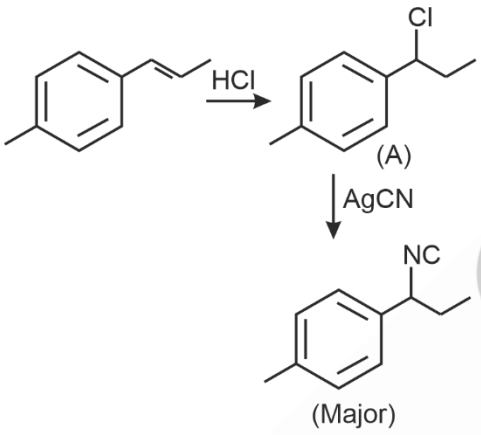


68. The product B formed in the following reaction sequence is :

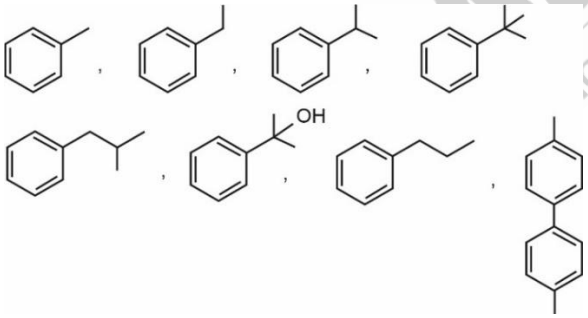


Answer (2)

Sol.



69. The total number of compounds from below when treated with hot $KMnO_4$ giving benzoic acid is :

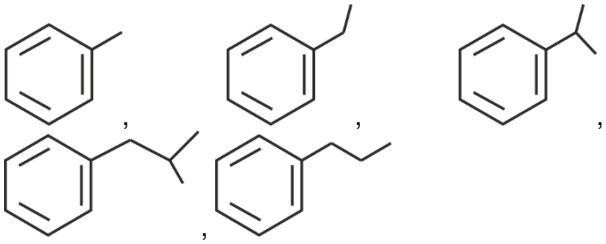


(1) 4 (2) 3
(3) 6 (4) 5

Answer (4)

Sol. The compounds having benzylic hydrogen will give benzoic acid on treatment with hot $KMnO_4$.

Which are following:



70. Match **List- I** with **List-II**.

	List-I (Saccharides)		**List-II (Glycosidic-linkages found)**
(A)	Sucrose	(I)	α 1 – 4
(B)	Maltose	(II)	α 1 – 4 and α 1 – 6
(C)	Lactose	(III)	α 1 – β 2
(D)	Amylopectin	(IV)	β 1 – 4

Choose the **correct** answer from the options given below:

(1) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)
(2) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
(3) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
(4) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

Answer (2)

Sol.

(A) Sucrose : $\alpha - C_1 - \beta - C_2$
(B) Maltose : $\alpha - C_1 - C_4$
(C) Lactose : $\beta - C_1 - C_4$
(D) Amylopectin : $\alpha - C_1 - C_4$ and $\alpha - C_1 - C_6$

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

71. Electrolysis of 600 mL aqueous solution of NaCl for 5 min changes the pH of the solution to 12. The current in Amperes used for the given electrolysis is _____. (Nearest integer).



Answer (2)

Sol. For electrolysis of $NaCl_{(aq)}$:

At cathode: $2H_2O + 2e^- \rightarrow H_2 + 2OH^-$

So, 2 moles of OH^- is produced by 2 F charge.

As pH = 12 $\Rightarrow$ $[OH^-] = 10^{-2}$

$$n_{OH^-} = \frac{10^{-2} \times 600}{1000} \text{ moles} = 6 \times 10^{-3} \text{ mol.}$$

6×10^{-3} moles of OH^- will be produced by 6×10^{-3} F charge

So, $i \times 5 \times 60 = 6 \times 10^{-3} \times 96500$

$$i = \frac{6 \times 96500}{5 \times 60 \times 1000} = 1.93 \simeq 2 \text{ A}$$

72. Total number of molecules/species from following which will be paramagnetic is _____.

$O_2, O_2^+, O_2^-, NO, NO_2, CO, K_2[NiCl_4], [Co(NH_3)_6]Cl_3, K_2[Ni(CN)_4]$

Answer (6)

Sol. Molecules/species having unpaired $e^{\ominus}$ are paramagnetic which are O_2, O_2^+, O_2^-, NO, NO_2, $K_2[NiCl_4]$

73. Consider the following data :

Heat of formation of $CO_2(g)$ = –393.5 kJ mol^{-1}

Heat of formation of $H_2O(l)$ = –286.0 kJ mol^{-1}

Heat of combustion of benzene = –3267.0 kJ mol^{-1}

The heat of formation of benzene is _____ kj mol^{-1}

(Nearest integer)

Answer (48)

Sol. (i) $C_{(S)} + O_{2(g)} \rightarrow CO_{2(g)}$, ΔH_1 = –393.5 kJ/mol

(ii) $H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(l)}$, ΔH_2 = –286.0 kJ/mol

(iii) $C_6H_6 + \frac{15}{2}O_{2(g)} \rightarrow 6CO_2 + 3H_2O + \Delta H_3$ = –3267.0 kJ/mol

(iv) $6C_{(s)} + 3H_{2(g)} \rightarrow C_6H_6$ ΔH_4 = ??

We can get the required eqn by (i) × 6 + (ii) × 3 – (iii)

So, $\Delta H_4 = 6 \times (-393.5) + 3 \times (-286.0) - (-3267.0)$

= 48 kJ/mol

74. The spin only magnetic moment (μ) value (B.M.) of the compound with strongest oxidising power among Mn_2O_3, TiO and VO is _____ B.M. (Nearest integer).

Answer (5)

Sol. Mn_2O_3 is strongest oxidising agent among the given.

$Mn_2O_3 : Mn^{3+} : [Ar]\ 4s^0 3d^4$

$$\mu = \sqrt{4(4+2)} = \sqrt{24} = 4.89$$

$\simeq 5$

75. A group 15 element forms $d\pi - d\pi$ bond with transition metals. It also forms hydride, which is a strongest base among the hydrides of other group members that form $d\pi - d\pi$ bond. The atomic number of the element is _____.

Answer (15)

Sol. Phosphorus of group is element forms hydride PH_3 which is strongest base among the hydrides of other group members that form $d\pi - d\pi$ bond.

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